

(c) Obtain the simplified functions in product of sums.

$$F_1$$

xz	00	01	11	10
0	0		0	
1		0		0

Simplified F_1 : Original F_1 of (a)

$$F_2$$

xz	00	01	11	10
0	0	0		0
1	0			

Simplified F_2 : $(y+z) \cdot (x+y) \cdot (xz)$

3-6: Obtain the simplified expressions in product of sums:

(a) $F(x,y,z) = \prod(0,1,4,5)$

Simplified F : y

$$F$$

xz	00	01	11	10
0	0	0		
1	0	0		

(b) $F(A,B,C,D) = \prod(0,1,2,3,4,10,11)$

Simplified F : $(B+C) \cdot (A+B) \cdot (A+C+D)$

$$F$$

AB	00	01	11	10
00	0	0	0	0
01	0			
11				
10			0	0

(c) $F(w,x,y,z) = \prod(1,3,5,7,13,15)$

Simplified F : $(w+\bar{z}) \cdot (\bar{x}+\bar{z})$

$$F$$

wz	00	01	11	10
00		0	0	
01		0	0	
11		0	0	
10				

3-7: Obtain the simplified expressions in (1) sum of products and (2) product of sums.